

Birla Institute of Technology and Science (BITS) Pilani (K. J. Somaiya)
Department of Electrical and Electronics Engineering
EEE E434 Digital Signal Processing
First Semester 2021-22

Date : 25/10/2021
Max. Time: 90 minutes

Mid-term

Weightage: 25 %
Max. Marks: 25 Marks

Instructions to the students:

1. Clearly show the procedure used to arrive at your answers for full credit.
2. Write solution of each question on fresh page.
3. Answer each subpart of a question in continuation.

LTI: Linear and Time Invariant, DFS: Discrete Fourier Series, DFT: Discrete Fourier Transform, DTFT: Discrete Time Fourier Transform, LPF: Low Pass Filter, ROC: Region of Convergence, IIT: Impulse Invariance Transformation, DE: Differential Equation, FFT: Fast Fourier Transform.

Questions 1 to 10 carry ONE mark each

1. The DTFT of $x[n]$ is given as $X(e^{j\omega}) = j \sin(\omega)[3 + 4 \cos(\omega) + 2 \cos^2(\omega)]$. Then find the $x[n]$.
2. The DFT of the two finite-length sequences $x_1[n]$ and $x_2[n]$ are given below
 $X_1[k] = \{4, 1 - j, -2, 1 + j\}$ and $X_2[k] = \{6, 1 - j, 0, 1 + j\}$
Find the circular convolution of $x_1[n]$ and $x_2[n]$.
3. The signal $x(t)$ is sampled with a sampling rate of 10 kHz. Let $X(k)$ is the 1000-point DFT of $x[n]$. Then find the continuous frequency (Hz) in $x(t)$, for an index value of $k = 200$ in $X(k)$.
4. The input to an LTI system is given by $e^{j\pi n}$. The impulse response of the system is given as $h[n] = \{0.3, 0, 2, 0.3\}$. Find the output of the system.
5. The phase response of a filter is given by $\theta(f) = -2\pi\alpha(f - f_c) - 2\pi\beta f_c$, Where f_c, α, β are constants. Then find the group delay and comment on the linear phase response of the filter.
6. Determine the poles of the Butterworth lowpass filter for the order $N=4$ (consider the cut-off frequency as 1 rad/sec)? Also sketch the location of poles on S-plane.
7. For the continuous transfer function, $H(s) = \frac{2}{s^2 + 3s + 2}$, Determine $H[z]$ using the IIT if the sampling interval is $T=1$ sec.
8. A causal LTI system is given by the DE $2y[n] = x[n] - x[n-1]$. Comment on the type of filter characteristics? (where $x[n]$ and $y[n]$ are input and output respectively).
9. What is the difference between DFS and DFT?
10. What are the ideal characteristics of a window function?

Questions 11 to 16 carry 2.5 marks each

11. Let $x[n]$ be a length 9 sequence given by $x[n] = \{3, 0, 1, -2, -3, 4, 1, 0, -1\}$. The DFT of $x[n]$ is $X(e^{j\omega})$. Then evaluate the following functions:

(a.) $X(e^{j0})$, (b.) $X(e^{j\pi})$, (c.) $\int_{-\pi}^{+\pi} X(e^{j\omega}) d\omega$, (d.) $\int_{-\pi}^{+\pi} |X(e^{j\omega})|^2 d\omega$, (e.) $\int_{-\pi}^{+\pi} \left| \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega$

12. A 498-point DFT $X[k]$ of a real valued sequence $x[n]$ has the following DFT samples:

$$X[0] = 2, X[111] = 7 + j3.1, X[a] = -2.2 - j1.5, X[112] = 3 - j0.7, X[b] = -4.7 + j1.9,$$

$$X[249] = 2.9, X[309] = -4.7 - j1.9, X[c] = 3 + j0.7, X[412] = -2.2 + j1.5, X[d] = 7 - j3.1.$$

Remaining DFT samples are assumed to be zero. Find the following:

(a.) Determine the values of the indices a, b, c, d .

(b.) Determine the expression for $x[n]$.

13. When the input to an LTI system is, $x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-n-1]$, the output is given as

$$y[n] = 6\left(\frac{1}{2}\right)^n u[n] - 6\left(\frac{3}{4}\right)^n u[n]. \text{ Then find the following:}$$

(a.) Find the system transfer function $H[z]$. Plot the pole and zeros of $H[z]$ and indicate the ROC.

(b.) Find the impulse response $h[n]$ of the system.

(c.) Comment on whether the system is stable and causal.

14. Two finite-length sequences $x_1[n]$ and $x_2[n]$ are zero outside the intervals $0 \leq n \leq 99$ and $10 \leq n \leq 39$, respectively. These two sequences are circularly convolved to obtain $y[n]$. Determine the set of values of n for $y[n]$ which is guaranteed to be identical to the linear convolution of $x_1[n]$ and $x_2[n]$.

15. Obtain the 8-point DFT of $x[n] = \{1, 1, 1, 0, 0, 0, 0, 0\}$ using DIT-FFT algorithm.

16. Design a Chebyshev Type-I digital IIR LPF using bilinear transformation by taking sampling period as 1 second to satisfy the following design specifications.

$$0.8 \leq |H(e^{j\omega})| \leq 1; \quad \text{for } 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.25; \quad \text{for } 0.6\pi \leq \omega \leq \pi$$

***** End of Paper*****